ANSWER SHEET DAC 2024

PRELIMINARY ROUND

Data Analysis Competition 2024

DATA ANALYSIS COMPETITION 2024

TEAM NAME: Dynamic Learners

TEAM ID: DAC135

CHAPTER I: Introduction

According to the International Atomic Energy Agency (IAEA), the demand for energy and electricity is continuously increasing and is expected to keep rising every year. The data also shows that electricity consumption is growing at a relatively high rate compared to overall energy consumption (see Appendix A.1). This indicates that electricity is a crucial sector in meeting future needs.

Electricity can be generated from various sources, including coal, gas, oil, nuclear, hydro, wind, bioenergy, and solar power. According to Our World in Data, the largest source of energy is coal, with 10,467.93 TWh consumed in 2023 [2]. Similarly, in Indonesia, coal is the largest source of electricity, reaching 216.78 TWh in 2023. However, as a non-renewable resource, coal's availability is threatened in the future. The burning of coal has negative environmental impacts, including greenhouse gas emissions and pollution, which can harm the environment, health, and climate.

To address these challenges, governments worldwide are increasingly considering the shift to cleaner and more sustainable energy sources, such as solar power. Solar energy offers an environmentally friendly solution with minimal emissions and negative impacts, and it holds the potential to meet future energy needs more sustainably. Indonesia, being situated on the equator, benefits from high solar energy potential with an average radiation of about 4.8 kWh/m² per day and an average of 12 hours of sunlight daily.

Choosing solar power for electricity generation has several advantages, including lower emissions and no fuel costs. Solar energy is a renewable resource with long-term availability. Using solar power can reduce electricity costs due to the absence of fuel expenses. Despite its many benefits, the uncertainty of solar energy production poses a challenge for companies in meeting consumer demand. Accurate predictions of solar energy output are essential for overcoming this challenge, allowing companies to plan strategies based on user needs. Factors such as wind, temperature, and humidity significantly affect solar energy production, making accurate data on these variables crucial. The main challenge is to optimize solar panel technology and improve solar energy production efficiency while considering environmental sustainability and efficient resource use. This report aims to evaluate the impact of environmental factors on solar energy production efficiency and develop strategies to optimize solar panel technology.

CHAPTER II: Theoretical Framework

A. Overview of Solar Energy Systems

The global electricity consumption continues to rise each year, driving the exploration of various alternative energy sources, one of which is solar power. Solar Power Plants (PLTS) are a promising solution for meeting environmentally friendly energy needs. Solar panels are the key components in PLTS, converting sunlight into electrical energy. Within solar panels are photovoltaic cells, which convert solar energy into electricity. Simply put, when photovoltaic cells absorb sunlight, electrons move between the positive and negative sides. This movement generates electrical energy that can be further utilized.

There are several types of solar panels, including Mono-Crystalline, Poly-Crystalline, and Thin Film Photovoltaic. Each type has different characteristics and energy efficiencies, which can be matched to specific needs. Choosing the right type of solar panel can impact the overall performance of the system. The effectiveness of solar panels is also influenced by various environmental factors. Sunlight intensity, weather conditions, clouds, humidity, temperature, and wind speed are some factors that can affect the amount of energy generated by solar panels.

B. K-Nearest Neighbor (KNN)

K-Nearest Neighbor (KNN) is a machine learning algorithm used for classification and regression. In this context, KNN is employed to address NaN (Not a Number) values in datasets by filling in missing values based on patterns found in existing data. The method works by finding a number of nearest data points (neighbors) within the feature space, and then determining the missing value by averaging the values of these neighbors. For example, in the "Hour" column, KNN searches for data points with similar time values to estimate the missing data.

KNN algorithm has emerged as a powerful tool in solar energy prediction and forecasting, demonstrating high accuracy and adaptability across various applications. Its effectiveness is particularly notable in assessing photovoltaic (PV) system performance and predicting solar radiation. KNN has been utilized to classify operational states of PV modules, achieving an accuracy of 99.2% in diagnosing conditions like partial shading and short circuits [8]. In a comparative study, KNN was one of the top-performing algorithms for predicting daily solar radiation, achieving R^2 values up to 99% [9].

C. Random Forest Regression

Random Forest Regression (RFR) is a powerful machine learning algorithm used for predictive modeling, particularly in scenarios where the relationship between input variables and the target variable is complex and non-linear. RFR operates by building multiple decision trees and combining their

outputs, which helps capture non-linear relationships and interactions between variables.

In the context of solar energy prediction, Random Forest Regression (RFR) has demonstrated its effectiveness by accurately modeling the relationship between various environmental factors and solar energy output. Research comparing different solar radiation prediction models has shown that Random Forest Regression is the most accurate, outperforming other models such as Multi-Layer Perceptron (NN-MLP) and Convolutional Neural Network (CNN) in terms of prediction accuracy [11]. Another study identified RFR as the most effective model for estimating solar power output, achieving an RMSE of 19.413 and an R^2 value of 0.968, indicating superior performance compared to other models like Support Vector Regression (SVR) and Multi-Layer Perceptron (MLP) [12].

D. Data Analysis Approach

1. Correlation Analysis
   Correlation analysis is a statistical method used to determine whether and how strongly pairs of variables are related. In this analysis, Pearson correlation is used to evaluate the relationship between environmental variables and solar energy production. This method measures the linear relationship between two continuous variables, with a range from -1 to +1. A value of +1 indicates a perfect positive linear relationship, -1 indicates a perfect negative linear relationship, and 0 indicates no linear relationship. The general formula for correlation can be found in Appendix A.3.
2. Feature Importance
   Feature importance refers to the quantification of how much individual features contribute to the predictions of a machine learning model. It indicates the extent to which altering a feature's value can influence the model's outcome, providing insights into the data-generating process (DGP). Various techniques exist to derive feature importance scores, each with different assumptions and interpretations. For this analysis, feature importance was determined using Random Forest, which assesses the contribution of each feature by analyzing the reduction in impurity achieved when features are used in decision-making within the model.
3. Root Mean Squared Error (RMSE)
   RMSE represents the magnitude of prediction error, where a smaller value (closer to 0) indicates higher prediction accuracy. The formula for calculating RMSE can be found in Appendix A.4.

CHAPTER III: Analytical Steps

A. Data Preparation

1. Data Source
   The data source for this analysis comes from the DAC Green Energy Company competition dataset, which includes measurements of environmental factors and solar energy production. This dataset encompasses variables that are measured periodically.
2. Merge Data
   Data merging is performed by linking datasets using the primary key Timestamp. In this case, Timestamp is used to align data from various sources at the same point in time to consolidate relevant information from different datasets. This ensures that each record is accurately connected based on time and facilitates a more comprehensive analysis.
3. Data Cleaning
   In this process, missing values are filled using the K-Nearest Neighbor (KNN) imputation method. KNN is employed to replace missing values in numeric columns by using the nearest values based on data similarity. KNN is chosen because it preserves the statistical structure of the data, works well with complex datasets, and is easy to apply to various variables, using 5 neighbors (k=5) for imputation in both the training and test datasets. Missing categorical data is filled with "No Data" to maintain data integrity.
4. Transform Data
   The cleaned data is transformed to match the Timestamp format required by the model and categorical variables are encoded into a suitable numeric format for the model.

B. Model Building

1. Training Model
   Random Forest Regression (RFR) was chosen for its ability to handle non-linear and complex relationships. The RFR model is trained using the prepared training data, with hyperparameters adjusted to achieve optimal performance. The dependent variable is `%Baseline` (the value to be predicted), while the independent variables include features such as weather and environmental data. One-Hot Encoding is applied to categorical variables to convert them into a numeric format and avoid multicollinearity by removing the first dummy column. The RFR model is created and trained on the training data using 100 decision trees (`n_estimators = 100`) for stability. The data is split into 80% for training and 20% for testing. After the model is trained on the training data, the test data is used to generate predictions for `%Baseline`.
2. Model Evaluation

Model evaluation is conducted using Root Mean Squared Error (RMSE) to measure the magnitude of prediction errors compared to actual values. The evaluation results show an RMSE of 0.079, indicating that the model has an average prediction error of 0.079. Additionally, feature importance is determined using RFR, which is calculated based on each feature's contribution to reducing impurity in the decision trees within the Random Forest. This helps identify the key factors influencing the predictions.

CHAPTER IV: Analysis of Results

Previously, data preprocessing and initial exploration were performed, including dataset cleaning, handling missing values, and visualizing the distribution of key variables. Here are the results of the analysis obtained:

A. Correlation Analysis

After processing the data, correlation analysis was conducted to understand the relationships between variables in the dataset and the predicted baseline. A correlation matrix and heatmap were used for visualization.

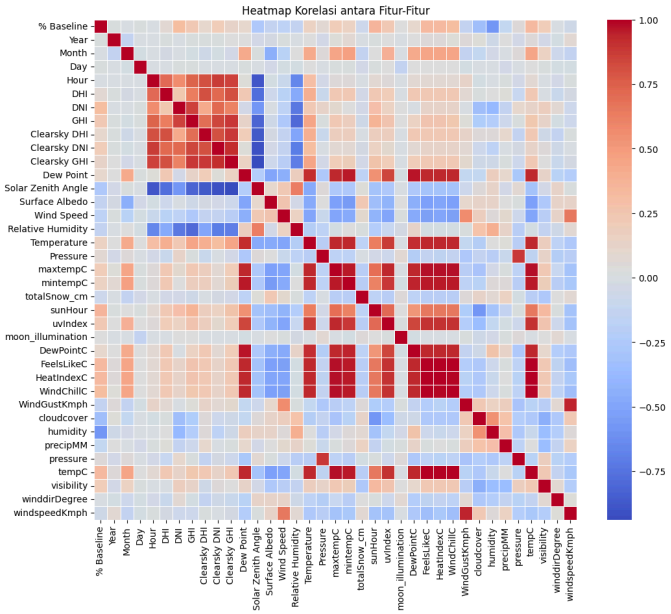

Figure 4.1. Correlation Heatmap between Features

Figure 4.1 shows a heatmap illustrating the strength of linear relationships between each pair of variables, with correlation values ranging from -1 to 1. Here are some variables that exhibit significant correlation with the baseline:

Variables with Positive Correlation:

1. SunHour (0.357586): Measures the number of sunlight hours received. A positive correlation of 0.357586 indicates that more sunlight hours

generally lead to higher energy production, as measured by % Baseline (see Appendix B.4).

2. HeatIndexC (0.328606): The heat index combines air temperature and humidity. The positive correlation suggests that an increase in the heat index tends to be associated with a higher % Baseline.
3. TempC (0.326254): Measures air temperature. The positive correlation indicates that higher temperatures generally support increased energy production. However, extremely high temperatures could reduce solar panel efficiency.
4. FeelsLikeC (0.319339): This accounts for the perceived temperature based on humidity and wind. The positive correlation with % Baseline suggests that higher perceived temperatures, similar to the heat index, can enhance energy production.
5. DNI (0.317226): Direct Normal Irradiance measures direct sunlight received by a surface perpendicular to the sun. The positive correlation shows that higher DNI levels are directly related to increased % Baseline (see Appendix B.5).

Variables with Negative Correlation:

1. Humidity (-0.591593): Measures air humidity. This significant negative correlation indicates that higher humidity is associated with a decrease in % Baseline (see Appendix B.1).
2. Cloudcover (-0.329698): Represents the percentage of cloud cover. The negative correlation suggests that increased cloud cover is associated with reduced energy production (see Appendix B.6).

The correlation analysis results show that some variables have significant linear relationships with the baseline and also with each other. These correlations may indicate dependencies or causal relationships that warrant further investigation, providing insights into how variables influence prediction outcomes.

B. Feature Importance

Correlation only measures linear relationships, whereas evaluating feature importance helps identify features with significant influence, including those with non-linear relationships. The Random Forest model is used because it can capture non-linear relationships and provide deeper insights into significant features.

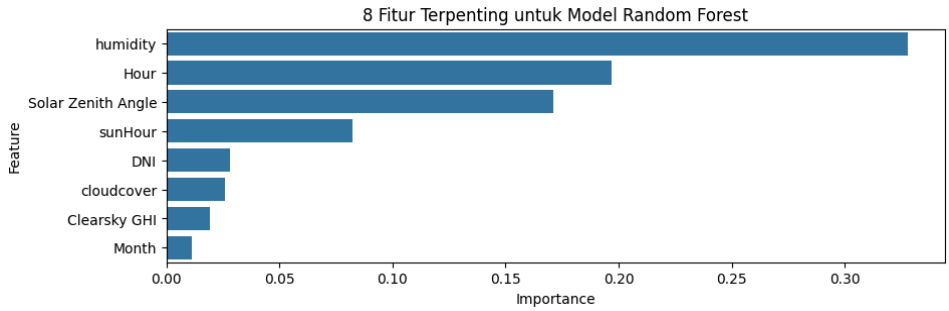

Figure 4.2. Key Features of the Random Forest Model

Figure 4.2 shows the features with the greatest impact on the prediction of %Baseline. Here are some of the most important features influencing the predictions based on the feature importance analysis:

1. Humidity (Importance: 0.327979): High humidity tends to reduce energy production efficiency because it can decrease the intensity of sunlight reaching the solar panels (see Appendix B.1).
2. Hour (Importance: 0.196772): The time of day is also a crucial factor. The graph shows that energy production generally increases at certain times, peaking around noon and then decreasing towards the evening. The most intensive sunlight occurs when the sun is at its highest point in the sky. The hourly average graph shows a curving pattern, with energy production peaking at specific hours and then declining (see Appendix B.2).
3. Solar Zenith Angle (Importance: 0.171099): This measures the angle between the sun and the vertical point at a specific location, with % Baseline tending to decrease as the zenith angle increases. The graph (see Appendix B.3) shows that the baseline is higher when the solar zenith angle is in the range of 60-80 degrees, with a curving pattern indicating a significant increase before a decrease. This might occur because a smaller solar zenith angle allows sunlight to fall more optimally on the solar panels, especially if the panels do not track the sun's angle precisely. Solar Zenith Angle also has a strong linear relationship with Hour, DHI, DNI, and GHI.
4. SunHour (Importance: 0.082185): The number of sunshine hours indicates the amount of direct sunlight available. More sunshine hours correlate with higher energy production potential (see Appendix B.4).
5. DNI (Importance: 0.028194): Direct Normal Irradiance is important because it provides optimal light intensity for solar panels. The percentage baseline tends to increase with higher DNI values (see Appendix B.5).
6. Cloudcover (Importance: 0.025867): More clouds block sunlight and reduce the efficiency of solar panels in generating energy (see Appendix B.6).

7. Clearsky GHI (Importance: 0.018996): Global Horizontal Irradiance measures sunlight when the sky is clear. Higher values generally indicate more energy for production. However, the graph (see Appendix B.7) shows that the relationship between Clearsky GHI and % Baseline tends to be non-linear. At low Clearsky GHI values, energy production is usually low, while at higher values, the median % Baseline varies. This indicates that other factors are influencing the results. Clearsky GHI has a strong linear relationship with other features like Hour, DHI, DNI, GHI, Clearsky DHI, Clearsky DNI, and Solar Zenith Angle.
8. Month (Importance: 0.011344): The month of the year affects energy production due to seasonal variations in sunlight availability. The monthly average graph (see Appendix B.8) shows a curving pattern, peaking around June to August. Energy production tends to be higher during these months.

C. Interpretation of Results and Conclusions

Based on the correlation analysis and feature importance, several key features have been identified as significantly influencing energy production. Features such as SunHour, HeatIndexC, TempC, FeelsLikeC, and DNI exhibit a strong positive correlation with % Baseline. Additionally, the time of day (Hour) and Solar Zenith Angle also play crucial roles in enhancing energy production efficiency.

Conversely, humidity and cloudcover show significant negative correlations. High humidity and extensive cloud cover contribute to reduced energy production by blocking the direct sunlight essential for optimal solar panel performance. Furthermore, some variables demonstrate strong correlations with other features, indicating potential interactions among these variables.

The Random Forest feature importance analysis reveals that humidity is the most influential feature, followed by Hour and Solar Zenith Angle. The Month feature indicates seasonal variations in energy production, peaking from June to August, with the highest daily production occurring around noon. While certain features show a correlation with the baseline, non-linear relationships and interactions between features also significantly impact energy production outcomes.

CHAPTER V: Conclusion and Recommendation

A. Summary and Conclusion

The correlation analysis revealed that `sunHour`, `HeatIndexC`, `tempC`, `FeelsLikeC`, and `DNI` have positive correlations with energy production. This indicates that more sunlight, higher temperatures, and increased heat index tend to

enhance energy production. Conversely, `humidity` and `cloudcover` show significant negative correlations, suggesting that higher humidity and greater cloud cover can reduce energy production.

Feature importance analysis also indicates that humidity, hour, and Solar Zenith Angle are the most influential factors in predicting the baseline. Other important factors include sunHour, DNI, cloudcover, Clearsky GH`, and month. This analysis identifies that air humidity, time of day, and solar zenith angle have the greatest impact on energy production, followed by sunlight duration, direct radiation, and seasonal factors. Enhancing energy production can be achieved by considering these factors in energy production planning and management.

B. Recommendation

Based on the analysis results, here are recommendations to improve solar energy production efficiency. These recommendations are based on the significant influence of various factors identified from the correlation and feature importance analysis:

1. Utilization of Sunlight Hours
   Optimizing the hours with the highest solar intensity, such as around midday (12 PM), proves effective for enhancing solar energy production. Analysis indicates that SunHour and Hour significantly impact energy production, with peak production occurring around noon when solar radiation is at its maximum. Scheduling operational times to align with these peak sunlight hours can maximize energy absorption and improve production outcomes.
2. Panel Angle Adjustment
   Adjusting solar panel angles to follow the Solar Zenith Angle is crucial for improving sunlight absorption efficiency. The Solar Zenith Angle has a significant feature importance value, highlighting its impact on energy production. Solar tracking technology, which automatically adjusts panel positions to follow the sun's movement, ensures that panels maintain an optimal angle throughout the day. This adjustment helps panels receive the maximum available sunlight, thereby enhancing system efficiency.
3. Humidity Management
   High humidity has a significant negative correlation with energy production. It can reduce energy production efficiency by obstructing sunlight absorption and degrading panel performance. Managing humidity around solar panels is essential. Measures such as reducing vegetation around panels and optimizing irrigation systems can help maintain humidity levels that do not affect panel performance. This will enhance

solar panel effectiveness and minimize potential production losses due to high humidity.

4. Cloud Cover Monitoring
   Cloud cover directly impacts solar energy production. Cloudcover shows a significant negative correlation with energy production, indicating that increased cloud cover can reduce production yields. Regular monitoring of cloud cover will allow for better operational adjustments. Companies should focus on panels in areas with lower cloud cover or adjust production schedules based on weather forecasts.
5. Weather Prediction Technology
   Integrating advanced weather prediction systems can help forecast optimal times for energy production. This is relevant as the Month feature shows seasonal variations in energy production, peaking during summer months such as June to August. Weather prediction systems can inform when to maximize energy production based on forecasted weather conditions.
6. Investment in Panel Materials
   Investing in solar panel materials that withstand high humidity and extreme temperatures can improve system durability and performance. TempC and HeatIndexC indicate that higher temperatures and heat indices positively correlate with energy production. Robust materials ensure that panels perform well under various environmental conditions, enhancing long-term production efficiency.

This report highlights key factors influencing solar energy production and proposes strategies to enhance system efficiency. Implementing these recommendations is expected to help DAC Green Energy Company optimize solar energy output and reduce uncertainties. The findings provide a strong foundation for developing more effective technologies and operational strategies, supporting the company's goals in renewable energy efficiency and sustainability.

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ATTACHMENT

Appendix A:

Figure A.1 : World Final Consumption of Energy and Electricity
Source: https://www-pub.iaea.org/MTCD/Publications/PDF/RDS-1-42_web.pdf

Final Consumption	2021	2030	2040	2050
Energy	422.4	471.2	498.3	535.1
Electricity	82.3	105.5	132.1	159.6
Electricity as % of Energy	*19.5%*	*22.4%*	*26.5%*	*29.8%*

Figure A.2: Electricity Production by Source
Source : https://ourworldindata.org/electricity-mix

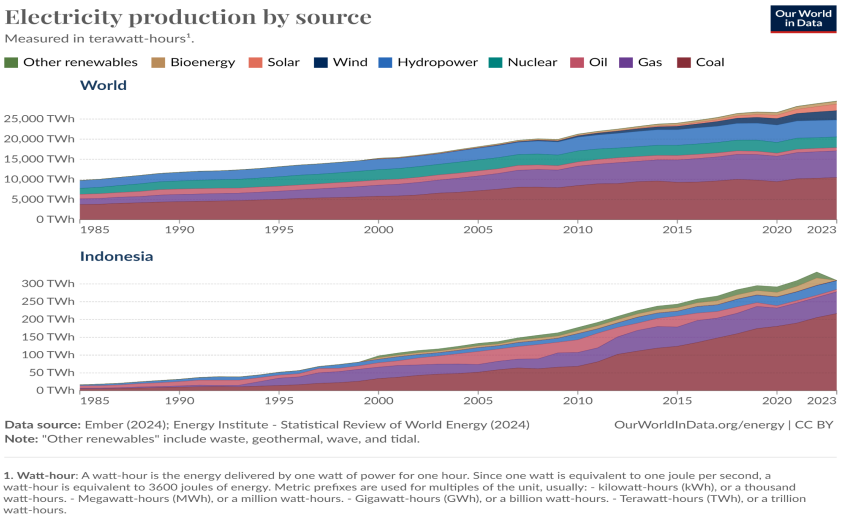

Figure A.3. Correlation Formula

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[\, n\sum x^2 - (\sum x)^2 \,][\, n\sum y^2 - (\sum y)^2 \,]}}$$

Table A.3. Meaning of the symbols

r	Coefficient Correlation
n	Number of data pairs
$\sum xy$	Sum of the product of each pair of x and y
$\sum x$	Total sum of the variable x

$\sum y$	Total sum of the variable y
$\sum x^2$	Sum of the squares of each value of x
$\sum y^2$	Sum of the squares of each value of y

Figure A.4. Root Mean Squared Error Formula

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

Table A.4. Meaning of the symbols

ŷi	Actual value
yi	Predicted value
n	Number of observations/rows

Appendix B:

Figure B.1: %Baseline pattern relative to Humidity

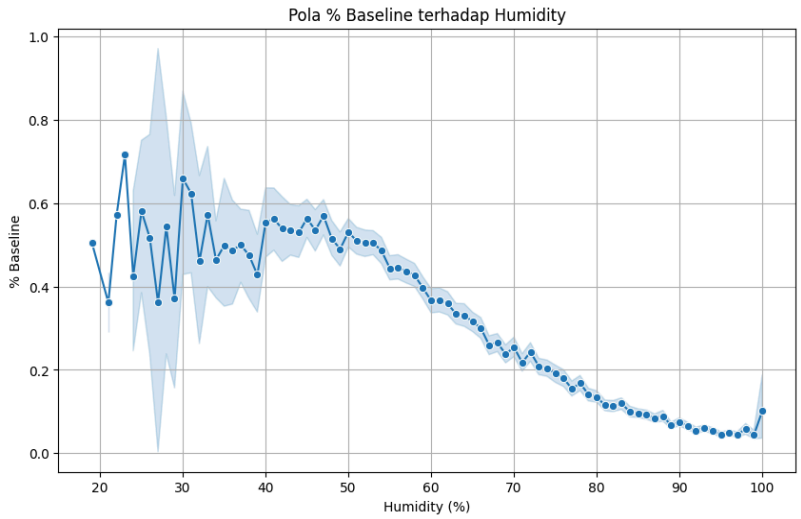

Figure B.2: Average %Baseline per hour

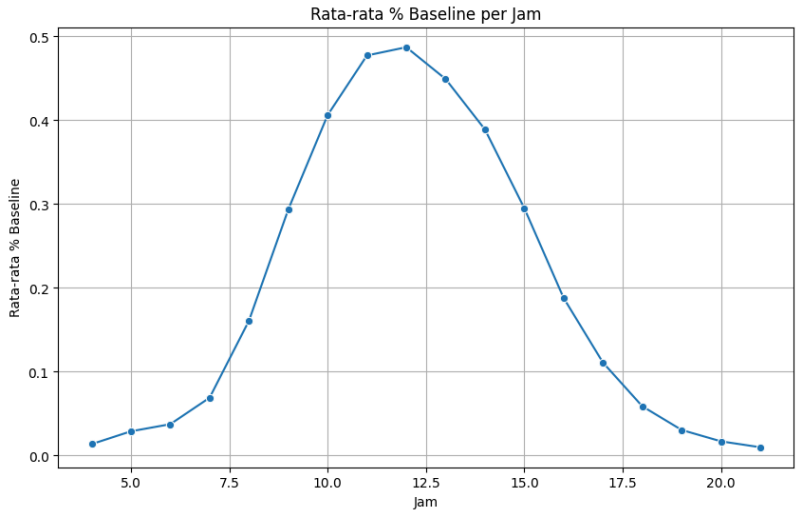

Figure B.3: The relationship between Solar Zenith Angle and %Baseline.

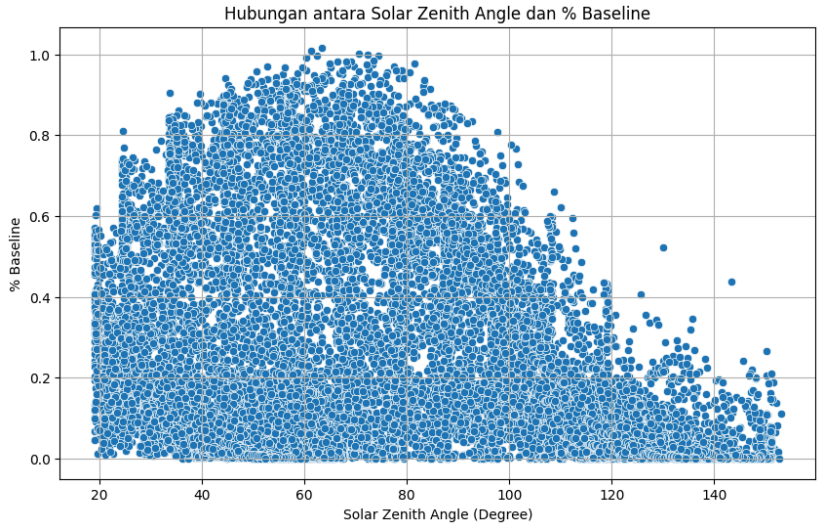

Figure B.4: The pattern of %Baseline in relation to sunHour

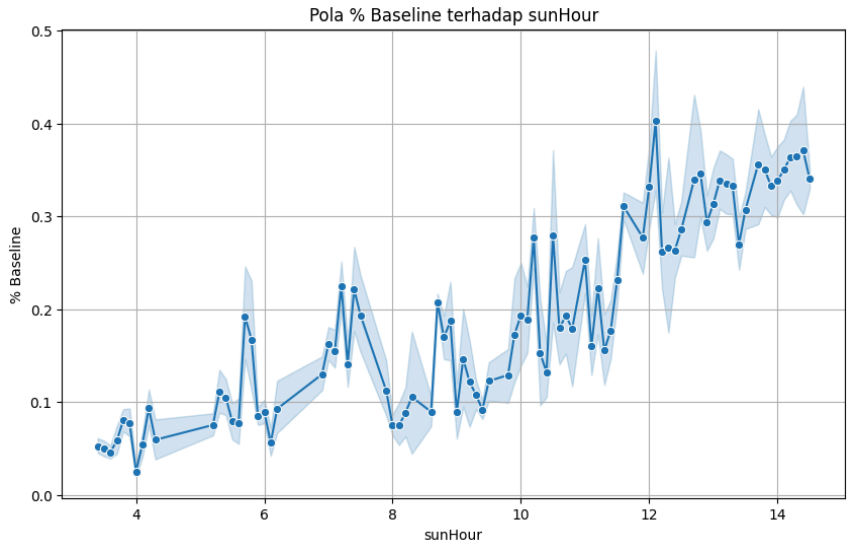

Figure B.5: The distribution of %Baseline based on DNI

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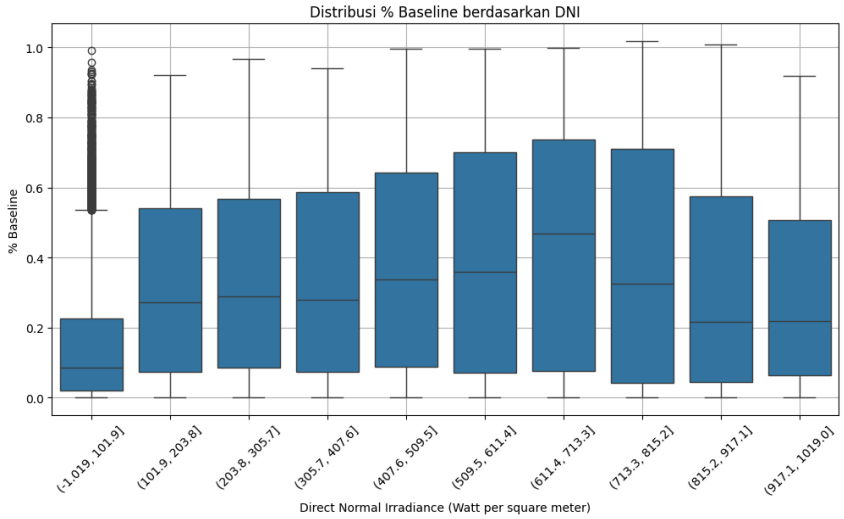


Figure B.6: The pattern of %Baseline in relation to cloud cover (%)

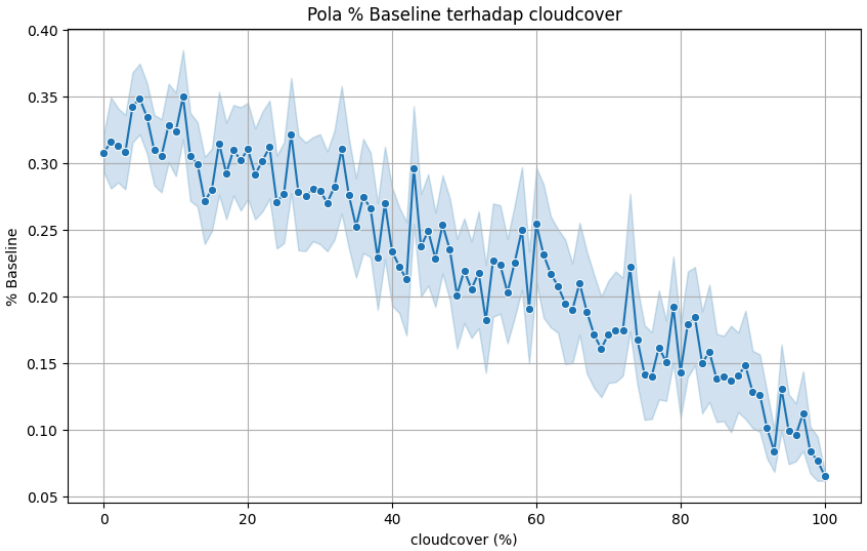


Figure B.7: Distribution of %Baseline based on Clearsky GHI

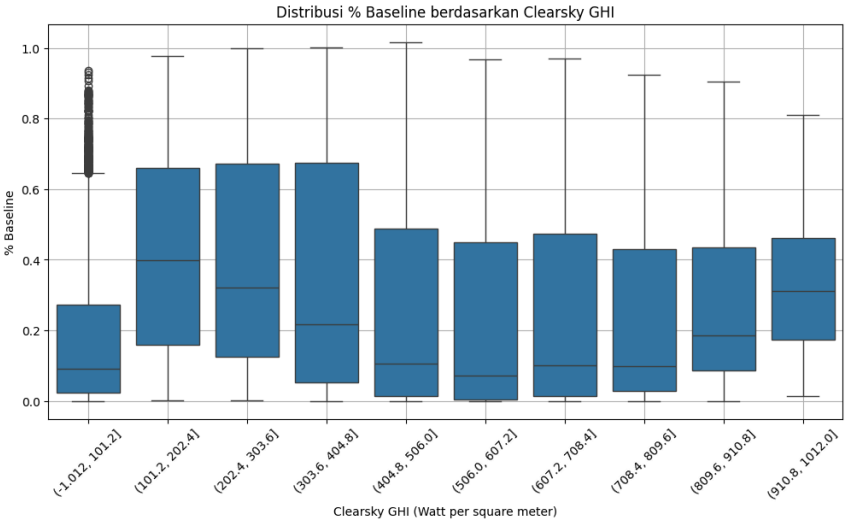
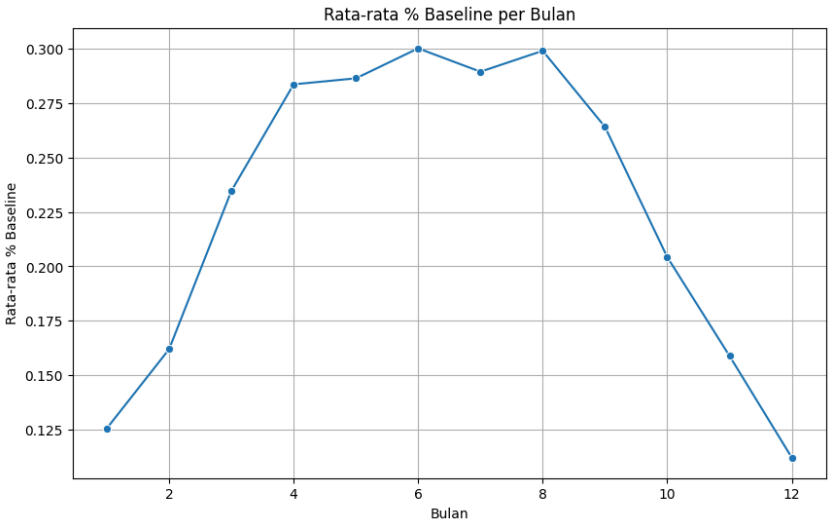


Figure B.8: The average %Baseline per month



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